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Studierichting: Informatica
Oefeningenreeks 3E nr. 14

$M(t) = 3t(10 - t)$

$M'(t) = 30 - 6t$

14.1

$M'(4) = 30 - 6 \cdot 4$

$= 30 - 24$

$\boxed{M'(4) = 6 \text{ g/u}}$

14.2

$\frac{dM}{dt} = 30 - 6t$

$\frac{5}{12} \text{ mg/s} = \frac{5}{12} \cdot \frac{3600}{1000} \text{ g/u}$

$\boxed{= \frac{3}{2} \text{ g/u}}$

$\frac{3}{2} = 30 - 6t$

$6t = 30 - \frac{3}{2}$

$6t = \frac{57}{2}$

$t = \frac{57}{12}$

$\boxed{t = \frac{19}{4} \text{ u}}$

14.3

$M'(t) = 0$

$30 - 6t = 0$

$6t = 30$

$\boxed{t = 5}$

14.4

t	0	5	$+\infty$
$M'(t)$	$+$	0	$-$
$M(t)$	$\nearrow$	$M(5)$	$\searrow$

De populatie is maximaal voor $t = 5$.

References